

Variable-naming pass — status

Goal: every C++ identifier in every module carries a name an interviewer would respect. Conventional names are kept (`i, j, k` indices; `n, m` sizes; `u, v` graph vertices; `lo, hi, mid`; `A, B, C` in a matrix-multiply context); cryptic ones are renamed.

Renames are applied to **code only** — the `// 1 x = L.head` pseudocode citations in comments are left intact, so each block still documents its correspondence with CLRS/Skiena.

Done (compiles clean under `g++ -std=c++17 -Wall -Wextra`)

MODULE HIGHLIGHTS OF THE RENAMING

M01	<code>nearestNeighborTour</code> , <code>closestPairTour</code> , <code>dist</code> → <code>euclidean</code> , 1-indexed merge/insertion sort
M02	selection sort, string match, matrix multiply, fast power, harmonic sums
M03	binary search, max-subarray, closest pair, bignum, Karatsuba, Strassen
M04	hiring problem, permute-by-sorting, reservoir sampling, <code>randomInt</code>
M05	heaps, quicksort/partition, counting/radix/bucket sort, <code>randomizedSelect</code>
M06	list nodes, sentinel vs plain list
M07	direct-address / chained / open-addressing tables, universal hash, Rabin-Karp, Bloom filter
M08	BST + red-black + order-statistic + interval + B-tree: <code>x/y/z/w/p/c/n/t</code> → <code>node</code> , <code>replacement</code> , <code>target</code> , <code>sibling</code> , <code>parent</code> , <code>children</code> , <code>keyCount</code> , <code>MinDegree</code> , <code>pivot/newParent</code> , <code>fullChild/newSibling</code>
M09	<code>s_</code> → <code>stack_</code> , <code>a_</code> → <code>bits_</code> , <code>v_</code> → <code>elements_</code> , <code>n_</code> → <code>count_</code> , merge helpers
M10	<code>p_</code> → <code>parent_</code> , <code>sets_</code> → <code>componentCount_</code> , <code>rx/ry</code> → <code>rootX/rootY</code> , <code>off_</code> → <code>offsetToParent_</code> , offline LCA / offline minimum
M11	DP tables named for what they hold: <code>r</code> → <code>bestRevenue</code> , <code>m</code> → <code>minCost</code> , <code>s</code> → <code>splitAt</code> , <code>c</code> → <code>lcsLen</code> , <code>D</code> → <code>dist</code> , <code>L</code> → <code>lisLenEndingAt</code> , <code>T</code> → <code>reachable</code> , <code>M</code> → <code>minMaxSum/chart</code> , <code>e/w/r</code> → <code>expectedCost/weight/rootCandidate</code>

M08's body and appendix were additionally re-verified behaviourally after the rename (randomized B-tree / red-black / order-statistic invariant checks) because the rotation and B-tree split were rewritten by hand rather than mechanically.

Remaining

MODULE	IDENTIFIERS TO FIX
M12 greedy	<code>a, t, x, ch, nd, idx, q, Q, h</code>
M13 traversal	<code>g, r, stk, q, c, d, pi</code> (keep <code>u/v</code>)
M14 MST	<code>e, g, r, x, b, a, c, p_</code>
M15 shortest paths	<code>r</code> (the <code>PathResult</code>), <code>d, g, pi, x, a</code>

Then: re-run the whole-tree compile + link check, rebuild the zip, re-sync to the project.

Tooling (session scratchpad, `bin/`)

- `rename.py FILE < map.json` — renames inside ``cpp blocks, code only, never in comments or string literals. Reports per-rule hit counts and flags no-op rules.
- `compile_md.py FILE` — extracts top-level cpp fences into a body TU and an appendix TU, compiles both with `-fsyntax-only`.
- `links.py DIR` — validates every markdown link and reports duplicate anchors.